

Wide-screen displays have an aspect ratio of 16:9. Many people are confused by the difference between 4:3 and 16:9, as at first glance they seem to be the same ratio. A simple math calculation will show you that $4:3 = 1.333$, whereas $16:9 = 1.777$. So it's quite clear that the 16:9 ratio means a much wider screen!

Most computer monitors are analogue display devices. This means that although the computer produces digital signals, the monitor only accepts analogue signals. The monitor connects to the video adapter or graphics card inside the computer to receive analogue signals.



The Graphics Card

This is the device that converts a computer's digital data output stream into analogue, so that the monitor can understand and display correctly. The graphics card receives digital data from the Central Processing Unit (CPU) and Random Access Memory (RAM) inside your computer, converts it all into an analogue signal and then sends the analogue signal to the monitor. This is what you finally see.

Most graphics cards also do advanced graphical computation and take away computing load from your CPU. These involve complex 3D computations that are required by most games of today.